

b) **Broadband:** includes several transmission technologies, such as:

- Digital Subscriber Line (DSL)
- Cable
- Wireless
- Satellite

1. **Digital Subscriber Line (DSL)** is a technology that provides a dedicated digital circuit between a residence and a telephone company's central office, allowing high-speed data transport over existing telephone lines. Typically, there is a modem on either end; one modem is located at the central office, and the other at the customer's site. There are several types of DSL including ADSL, HDSL, IDSL, SDSL, and VDSL, but of these, the most common, at least in the UK and India is ADSL.

ADSL (Asymmetric Digital Subscriber Line) is designed to deliver more bandwidth downstream (from the central office to the customer site) than upstream (that's why the word "asymmetric").

SDSL, common in Europe, is the *Symmetric Digital Subscriber Line*, a technology that allows the same data rates for upstream as well as downstream traffic, over existing copper telephone lines. However, it doesn't allow simultaneously voice connections on the same wires.

2. Cable:

As millions of homes already have cable TV connections, cable TV service providers simply equip the customer with a cable modem and provide access to the Internet over cable TV lines.

The coaxial cable allows for much greater bandwidth than telephone lines, so a cable modem is considerably faster than DSL (about 3 Mbps).

3. Wireless Internet Access

Wireless communication is nothing but the transferring of data over a particular distance without using the long, irritating, tubular things called wires.

Certain ISPs offer access to the Internet without your first having to plug in a wire to a jack. Such services may use different types of wireless technologies: